Classic 5.8 GHz analog VTX uses wideband FM for video and adds an audio subcarrier (commonly ~6.5 MHz) FM–modulated by your mic line. Your cam's AUDIO pin (or an external mic board) goes to VTX AUDIO IN through an AC–coupling cap (1 µF–4.7 µF), sometimes with a small series resistor (e.g., 1 k–10 k) per the VTX datasheet. (This mirrors TV subcarrier practice; exact frequency is VTX–specific.)

Analog VRX – ??

ANTENNA DO NOT FORGET

DONT FORGET COUPLING AND DECOUPLING CAPS ON INPUTS AND OUTPUTS RF

RX5808 (common in DIY ground stations and goggles).
RC832 or similar board–level modules.
Check youtube
Coming soon in openpixel
RTC6715
STM32F103C8T6 or STM32F3
https://github.com/goebish/triversity-multipath-erazer
Mechanical Mounting Hole
SCREEN + AUDIO MCU
SCREEN MCU
tinyFINITY (Need tinyOSD)
Or RaspberryPi Based
Or ESP based
Antenna > Filter > RTC6715 > ARDUINO > RTC6715 > Screen
Arduino Control Freq. Bands only!
Screen Connection
AV (composite) input CVBS SCREEN
Depends if SPI TFT or I2C OLED or lcd
Speaker Connection
Directly into screen audio IN Or I2S?
RTC6715 > POT > DClass AMP > Speaker
Ground stations can increase range!
AMP LNA
NEED 1.2V Generator
AMT630A
Mech GND
Touch X1,Y1,X2,Y2
AT043TN24
AT070TN92
Hirose FH19SC–40S–0.5SH
54132–4062
U2
RTC6715
3V3_2A_VRX

For next edition place decouple caps close to pins properly

Do you want to build FPV monitor?
Vitaliy

– 17:05
In this sequence there is not enough RAM to store the buffer with ADV7180... external memory is needed
Vitaliy

– 17:29
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If you need your own firmware for FPV monitor, contact me and I will tell you how.

Awesome—let's lock it in for AT043TN24 V.4 (backlight = 9 LEDs in series, 28.8 V typ / 31.5 V max @ 20 mA).

TPS61169 setup (values you can place)

VIN (to TPS61169): 3.3 V (your schematic's 3.3V_A rail).

I_LED (backlight current): 20 mA → RSET = 0.204 V / 0.02 A = 10.2 Ω (FB→GND).

Inductor (L): 4.7 µH, Isat ≥ 1 A (good margin; e.g., Coilcraft LPS4018–472ML, Isat≈1.9 A).

Schottky diode (D): Use ≥ 60 V reverse rating (e.g., SS16 / MBR0560).
Why 60 V? Your string is ~29-31.5 V and the IC's open–LED protection can push VOUT toward ~37-39 V; 60 V gives safe headroom (40 V parts like SS14 are marginal here).

Input cap (CIN): 4.7 µF, ≥ 6.3 V X5R/X7R, close to VIN/GND pins.

Output cap (COUT): 2.2-4.7 µF, ≥ 50 V X5R/X7R, close to LED+ return node. (50 V covers 31.5 V max plus spikes.)

CTRL (PWM dimming): Drive from your AMT630A B/L_EN/PWM pin (5-100 kHz).

Layout: Keep SW-L-D-COUT loop short and fat; place CIN tight to VIN/GND; put RSET right at FB.

What voltages/currents you'll see

VOUT (LED+): Set by LED string → ≈ 28.8 V typ, ≈ 31.5 V max (TPS61169 can go up to ~38-39 V with OVP).

I_LED (through backlight): 20 mA (fixed by RSET=10.2 Ω).

IIN (from 3.3 V): ≈ 0.21 A at 28.8 V/20 mA (assuming ~85% efficiency).
Calculation:

≈

.

.

⇒
28.8
·
0.02
0.85
·
3.3
≈
0.205

A
I
IN

≈
η·V
IN

V
OUT

·I
LED

⇒
0.85·3.3
28.8·0.02

≈0.205 A. (Budget 200-220 mA.)

Switch/inductor peak current (check): with L=4.7 µH, f≈1.2 MHz, worst case near OVP, ≈ 0.55 A peak—well below the TPS61169 min 1.2 A current limit and fine for a ≥1 A–Isat inductor.

Hook–up (matches your schematic)

SW → inductor → VIN, Schottky from SW/inductor node to LED+; LED– → FB → RSET → GND; CTRL from B/L_EN. (Exactly the same block where your KA2708 sat.)

If you want, share the exact panel datasheet page for the LED spec it lists; I'll sanity–check the RSET, finalize COUT voltage, and suggest specific Mouser part numbers for L/D/C that drop straight in.

IFOUT – Analog Out. 480 MHz IF output from the mixer/IF front–end; normally routed out to an external SAW band–pass filter and back into IFIN.
opendevices.ru

IFIN – Analog In. Return from the SAW filter into the chip's IF amplifier/demod chain. Keep the IFOUT-SAW-IFIN path very short/controlled.
opendevices.ru

BBOUT2 – Analog Out. FM video baseband (demod) buffer output. Typically AC–coupled and fed to the internal video amp input (VAMPIN), with optional external RC shaping.
opendevices.ru

VAMPIN – Analog In. Video amplifier input (takes the demodulated video from BBOUT2).
opendevices.ru

VAMPOUT – Analog Out. Video amplifier output; drive your CVBS output here (AC–couple and terminate for ~1 Vpp into 75 Ω).
opendevices.ru

FMIN_65 – Analog In. 6.5 MHz audio subcarrier input to the audio demod path; normally taken from the video baseband via a 6.5 MHz ceramic band–pass filter placed between FMIN_65 and AUIN_65.
opendevices.ru
+1

AUIN_65 – Analog In. Input to the 6.5 MHz audio demod/amp after the ceramic filter.
opendevices.ru

AUOUT_65 – Analog Out. Demodulated audio output from the 6.5 MHz path (line–level; AC–couple).
opendevices.ru

VT_65 – Analog Out. 6.5 MHz audio demodulator VCO/PLL loop node ("audio demod out" in table). You add the small RC loop–filter network here; community docs show a working network (e.g., ~200 Ω/430 k/470 pF/100 pF + 4.7 µF).
opendevices.ru
+1

AGC_C – Analog I/O. AGC rectifier/output node for the IF chain; decouple with a capacitor to set AGC dynamics. You can also monitor it as an AGC indicator. (For a calibrated strength output, use RSSI pin IFPDC_C.)

CVBS → ADV7180/ADV7280/TVP5150 (BT.656) →

STM32F7/H7 (DCMI + LTDC) → RGB TFT, or

ESP32–S3 (parallel in + RGB LCD) for modest resolutions, or

Raspberry Pi (Zero 2/4) via USB AV capture → HDMI/DSI.

Audio: RTC6715 AFOUT → I²S codec/ADC → SoC, or bypass SoC and go straight to an amp.

Practical, safe rules

Within the RGB bus (R/G/B data lines): keep trace lengths matched within ±10 mm (±0.4").

Data vs DCLK (the only one that really matters): keep each data line within ±20 mm of DCLK length.

HS/VS/DE vs DCLK: treat like data; ±20 mm is plenty.

I got this so far is this correct do you think?

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Others input is also appreciated!!!
Image
Image

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And for dealing with audio and video, what should be best? i was thinking a class D amp for audio but for the video processing so i can output on a common oled or tft lcd not sure?

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Thanks for that, we have worked with the VM/MM238R from SUNFLYTEK (CHN) as well. Similar units and found in most of the analog goggles modules out there.